
1 Objective(s)

- To learn typesetting of journal articles, technical reports, books, and slide presentations.
- To learn formatting documents containing sectioning, cross references, tables, equations and figures.
- To learn typesetting of complex mathematical formulae.

2 LATEX

LATEX is used all over the world for scientific documents, books, as well as many other forms of publishing. Not only can it create beautifully typeset documents, but it allows users to very quickly tackle the more complicated parts of typesetting, such as inputting mathematics, creating tables of contents, referencing and creating bibliographies, and having a consistent layout across all sections. Due to the huge number of open source packages available (more on this later), the possibilities with LATEX are endless. These packages allow users to do even more with LATEX, such as add footnotes, draw schematics, create tables etc.

One of the most important reasons people use LATEX is that it separates the content of the document from the style. This means that once you have written the content of your document, we can change its appearance with ease. Similarly, you can create one style of document which can be used to standardise the appearance of many different documents. This allows scientific journals to create templates for submissions. These templates have a pre-made layout meaning that only the content needs to be added. In fact there are hundreds of templates available for everything from CVs to slideshows.

3 Getting Started with LATEX

https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes#What_is_LaTeX.3F

Title and Abstract

- ▶ Tell L^AT_EX the `\title` and `\author` names in the preamble.
- ▶ Then use `\maketitle` in the document to actually create the title.
- ▶ Use the `abstract` environment to make an abstract.

```
\documentclass{article}

\title{The Title}

\author{A. Author}

\date{\today}

\begin{document}
\maketitle

\begin{abstract}
Abstract goes here...
\end{abstract}

\end{document}
```

The Title
A. Author
January 26, 2020

Abstract
Abstract goes here...

Sections

- ▶ Just use `\section` and `\subsection`.
- ▶ Can you guess what `\section*` and `\subsection*` do?

```
\documentclass{article}
\begin{document}

\section{Introduction}

The problem of \ldots

\section{Method}

We investigate \ldots

\subsection{Sample Preparation}

\subsection{Data Collection}

\section{Results}

\section{Conclusion}

\end{document}
```

1 Introduction

The problem of ...

2 Method

We investigate ...

2.1 Sample Preparation

2.2 Data Collection

3 Results

4 Conclusion

Labels and Cross-References

- ▶ Use `\label` and `\ref` for automatic numbering.
- ▶ The `amsmath` package provides `\eqref` for referencing equations.

```
\documentclass{article}
\usepackage{amsmath} % for \eqref
\begin{document}

\section{Introduction}
\label{sec:intro}

In Section \ref{sec:method}, we \ldots

\section{Method}
\label{sec:method}

\begin{equation}
\label{eq:euler}
e^{i\pi} + 1 = 0
\end{equation}

By \eqref{eq:euler}, we have \ldots

\end{document}
```

1 Introduction

In Section 2, we ...

2 Method

$$e^{i\pi} + 1 = 0$$

(1)

By (1), we have ...

Structured Documents Exercise

Typeset this short paper in $\text{\LaTeX}$: ¹

Click to open the paper

Make your paper look like this one. Use $\text{\ref}$ and $\text{\eqref}$ to avoid explicitly writing section and equation numbers into the text.

Click to open this exercise in **Overleaf**

- Once you've tried, [click here to see my solution](#).

¹From <http://pdos.csail.mit.edu/scigen/>, a random paper generator.

Graphics

- Requires the `graphicx` package, which provides the $\text{\includegraphics}$ command.
- Supported graphics formats include JPEG, PNG and PDF (usually).

```
\includegraphics[
  width=0.5\textwidth]{gerbil}

\includegraphics[
  width=0.3\textwidth,
  angle=270]{gerbil}
```

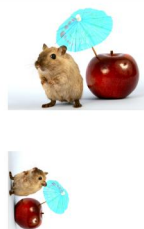


Image license: CC0

Interlude: Optional Arguments

- ▶ We use square brackets `[ ]` for optional arguments, instead of braces `{ }`.
- ▶ `\includegraphics` accepts optional arguments that allow you to transform the image when it is included. For example, `width=0.3\textwidth` makes the image take up 30% of the width of the surrounding text (`\textwidth`).
- ▶ `\documentclass` accepts optional arguments, too. Example:
`\documentclass[12pt,twocolumn]{article}`

makes the text bigger (12pt) and puts it into two columns.

- ▶ Where do you find out about these? See the slides at the end of this presentation for links to more information.

Floats

- ▶ Allow $\text{\LaTeX}$ to decide where the figure will go (it can “float”).
- ▶ You can also give the figure a caption, which can be referenced with `\ref`.

```
\documentclass{article}
\usepackage{graphicx}
\begin{document}

Figure \ref{fig:gerbil} shows \ldots

\begin{figure}
\centering
\includegraphics[%
width=0.5\textwidth]{gerbil}
\caption{\label{fig:gerbil}Aww\ldots.}
\end{figure}

\end{document}
```

Image license: CC0



Figure 1: Aww....

Figure 1 shows ...

Tables

- ▶ Tables in $\text{\LaTeX}$ take some getting used to.
- ▶ Use the `tabular` environment from the `tabularx` package.
- ▶ The argument specifies column alignment — `left`, `right`, `right`.

<pre>\begin{tabular}{lrr} Item & Qty & Unit \\\\$ \\\ Widget & 1 & 199.99 \\\ Gadget & 2 & 399.99 \\\ Cable & 3 & 19.99 \\\ \end{tabular}</pre>	<table><tr><td>Item</td><td>Qty</td><td>Unit \$</td></tr><tr><td>Widget</td><td>1</td><td>199.99</td></tr><tr><td>Gadget</td><td>2</td><td>399.99</td></tr><tr><td>Cable</td><td>3</td><td>19.99</td></tr></table>	Item	Qty	Unit \$	Widget	1	199.99	Gadget	2	399.99	Cable	3	19.99
Item	Qty	Unit \$											
Widget	1	199.99											
Gadget	2	399.99											
Cable	3	19.99											

- ▶ It also specifies vertical lines; use `\hline` for horizontal lines.

```

\begin{tabular}{|l|rr|} \hline
Item & Qty & Unit \\\$ \\\hline
Widget & 1 & 199.99 \\\
Gadget & 2 & 399.99 \\\
Cable & 3 & 19.99 \\\hline
\end{tabular}

```

Item	Qty	Unit \$
Widget	1	199.99
Gadget	2	399.99
Cable	3	19.99

- ▶ Use an ampersand `&` to separate columns and a double backslash `\\` to start a new row (like in the `align*` environment that we saw in part 1).

bib $\text{\TeX}$ 1

- ▶ Put your references in a `.bib` file in 'bibtex' database format:

```
@Article{Jacobson1999Towards,
  author = {Van Jacobson},
  title = {Towards the Analysis of Massive Multiplayer Online
    Role-Playing Games},
  journal = {Journal of Ubiquitous Information},
  Month = jun,
  Year = 1999,
  Volume = 6,
  Pages = {75--83}}

@InProceedings{Brooks1997Methodology,
  author = {Fredrick P. Brooks and John Kubiawicz and
    Christos Papadimitriou},
  title = {A Methodology for the Study of the
    Location-Identity Split},
  booktitle = {Proceedings of OOPSLA},
  Month = jun,
  Year = 1997}
```

- ▶ Most reference managers can export to bibtex format.

bibT_EX 2

- ▶ Each entry in the .bib file has a *key* that you can use to reference it in the document. For example, Jacobson1999Towards is the key for this article:

```
@Article{Jacobson1999Towards,  
  author = {Van Jacobson},  
  ...  
}
```

- ▶ It's a good idea to use a key based on the name, year and title.
- ▶ L^AT_EX can automatically format your in-text citations and generate a list of references; it knows most standard styles, and you can design your own.

bibT_EX 3

- ▶ Use the natbib package² with `\citet` and `\citep`.
- ▶ Reference `\bibliography` at the end, and specify a `\bibliographystyle`.

```
\documentclass{article}  
\usepackage{natbib}  
\begin{document}  
  
\citet{Brooks1997Methodology}  
show that \ldots. Clearly,  
all odd numbers are prime  
\citep{Jacobson1999Towards}.  
  
\bibliography{bib-example}  
% if 'bib-example' is the name of  
% your bib file  
  
\bibliographystyle{plainnat}  
% try changing to abbrnat  
  
\end{document}
```

Brooks et al. [1997] show that Clearly, all odd numbers are pri
[Jacobson, 1999].

References

Frederick P. Brooks, John Kubiatowicz, and Christos Papadimitriou. A method
along for the study of the location-identity split. In *Proceedings of OOPSL*.
June 1997.

Van Jacobson. Towards the analysis of massive multiplayer online role-play
games. *Journal of Ubiquitous Information*, 6:75–83, June 1999.

²There is a new package with more features named bibl_atex but most of the articles templates still use natbib.

4 Lab Task (Please implement yourself and show the output to the instructor)

1. Show the output learnt from the link to the Course instructor.

5 Policy

Copying from internet, classmate, seniors, or from any other source is strongly prohibited. 100% marks will be *deducted* if any such copying is detected.

Authors in [Kailash Varma et al., 2024] said "this". another sentence Kailash Varma et al. [2024]. Singh et al. [2021b] and another paper got referenced Singh et al. [2021a]

References

- Nadimpalli Madana Kailash Varma, Md. Irfan Ahmed, G. Madhusudhan, G. Rishab Babu, T. Shalini, and Fatima Unnisa. Machine learning-enabled smart transit: Real-time bus tracking system for enhanced urban mobility. In *2024 Intelligent Systems and Machine Learning Conference (ISML)*, pages 520–524, 2024. doi: 10.1109/ISML60050.2024.11007439.
- Soni Singh, K R Ramkumar, and Ashima Kukkar. Machine learning techniques and implementation of different ml algorithms. In *2021 2nd Global Conference for Advancement in Technology (GCAT)*, pages 1–6, 2021a. doi: 10.1109/GCAT52182.2021.9586806.
- Soni Singh, K R Ramkumar, and Ashima Kukkar. Machine learning techniques and implementation of different ml algorithms. In *2021 2nd Global Conference for Advancement in Technology (GCAT)*, pages 1–6, 2021b. doi: 10.1109/GCAT52182.2021.9586806.